Various notes on MLS Chapter 2, Section 5

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On the dot products on p. 61:

Just as MLS often write the cross product between two vectors, where the vectors are expressed in component form as column vectors, as simply the first column vector $\times$ the second, they also often do this with dot products, like on p. 61. In my view, a cleaner and often less confusing way, of writing the dot product in terms of vectors expressed in component form in column vectors, for instance the one in the integral equation at the bottom of the page, is

$$W = \int_{t_1}^{t_2} V_{ab}^{bT} F_b \, dt$$

The derivation of eq. (2.65) p. 62

Since frames B and C are fixed to a single rigid body, $\dot{g}_{bc}$ is a zero matrix. By applying eq. (2.53) p. 54, $V_{bc}^s = \left(\dot{g}_{bc} g_{bc}^{-1}\right)^\vee = 0$, a 6×1 zero vector. Therefore, by proposition 2.14 p. 59, $V_{ac}^s = V_{ab}^s$. Similarly, by applying eq. (2.55) p. 55, $V_{bc}^b = \left(g_{bc}^{-1} \dot{g}_{bc}\right)^\vee = 0$, also 6×1 zero vector. Therefore, by proposition 2.15 p. 59, $V_{ac}^b = \mathrm{Ad}_{g_{bc}^{-1}} V_{ab}^b$ and thus $V_{ab}^b = \mathrm{Ad}_{g_{bc}} V_{ac}^b$.

Using this, a cleaner way of writing the unnumbered equation just before eq. (2.65) is

$$V_{ac}^{bT} F_c = V_{ab}^{bT} F_b = \left(\mathrm{Ad}_{g_{bc}} V_{ac}^b\right)^T F_b = V_{ac}^{bT} \mathrm{Ad}_{g_{bc}}^T F_b \ .$$

From this follows eq. (2.65).

For the last equation before Example 2.7 p. 63

$$V^s = \mathrm{Ad}_g V^b \qquad (2.61)$$

$$F^b = \mathrm{Ad}_g^T F^s \qquad (2.67)$$

From eq. (2.61) follows

$$V^b = \mathrm{Ad}_g^{-1} V^s$$

Therefore

$$\delta W = V^{bT} F^b = V^{sT} \mathrm{Ad}_g^{-T} \mathrm{Ad}_g^T F^s = V^{sT} F^s \ .$$

Eq. (2.68) p. 64

$$\begin{aligned} F &= M \begin{bmatrix} \omega \\ -\hat{\omega} q + h\omega \end{bmatrix} && h \text{ finite} \\ F &= M \begin{bmatrix} 0 \\ \omega \end{bmatrix} && h = \infty \end{aligned}$$

It is worth noting that, for the case where h is finite, the moment of the offset force $M\omega$ about the origin of A is a cross product of the form $\vec{r} \times \vec{F}$ (as derived in the Solid Mechanics 1 course), which here can be expressed as $\hat{q}\omega M = -\hat{\omega} q M$. This is mentioned by MLS in the text.

Proof of Theorem 2.17 (Poinsot) Case 2, p. 65

To proof that the suggested solution, one of many, is in fact a solution, it is substituted into the torque equation, that is,

$$h = \frac{f^T \tau}{||f||^2} \qquad \text{and} \qquad q = \frac{\hat{f} \tau}{||f||^2}$$

are substituted into

$$M(\hat{q}\omega + h\omega) = \tau \ .$$

The left hand side of this equation, with the substitution, and realizing that $||f|| = M$, renders:

$$M \left[\widehat{\left(\frac{\hat{f}\tau}{M^2} \right)} \omega + \frac{f^T\tau}{M^2}\omega \right] \ .$$

Now taking into account that $\omega = f/M$ changes the above expression of the left hand side to

$$\widehat{(\hat{\omega}\tau)}\omega + \left(\omega^T\tau\right)\omega$$

This expression is equal to τ when $||\omega|| = 1$, which it is. It is an identity that we have in fact used earlier, in the discussion of MLS section 3.3 "Screws: a geometric description of twists" (see the slides set under March 16 on Moodle). This identity is proven in the MATLAB script "ident4MLS Theor2 17.pdf", which is also available on Moodle under the "Notes" topic.

So, in our proof above we have shown that for the chosen expressions of h and q, the left hand side reduces to τ and is therefore indeed equal to the right hand side; in other words that the suggested solution renders a valid equation and thus is a valid solution.

Clarification of Poinsot's theorem Case 2, p 65

In the statement of Poinsot's theorem reference is made to "…a torque about the same axis." In the case where $f \neq 0$ (case 2), one should understand this to refer to the compoment $Mh\omega$ in the expression for the torque

$$\tau = M\hat{q}\omega + Mh\omega \ ,$$

that is, this statement should not be understood to mean that the vector τ is in the same direction as f. One can indeed interpret the above expression of τ as that the vector τ can be decomposed into two components:

- $Mh\omega$ is the component of τ in the direction of f. Note that, since $M = ||f||$ and $h = f^T\tau/||f||^2 = f^T\tau/M^2$, one can say that $Mh = \omega^T\tau$, that is, Mh is the dot product of τ with the unit vector in the direction of f. Therefore, Mh is the component of τ in the directrion of f, as a scalar. This same component, expressed as a vector, is therefore $Mh\omega$.
- $M\hat{q}\omega$ is the component of τ normal to the vector f. To show that $M\hat{q}\omega$ is normal to f, we can calculate its dot product with ω: $\omega^T M\hat{q}\omega = -M\omega^T\hat{\omega}q = M\omega^T\hat{\omega}^T q = Mq^T\hat{\omega}\omega = 0$, which proves the statement.

Correction of text, 1st paragraph after definition 2.3 p. 66:

When the authors describe screw S_i, stating that it has an axis l_i, they do so in terms of, among others, a parameter $\lambda \in \mathbb{R}^1$. This parameter, however, is not necessarily the same for all screws. It is therefore more appropriate to also have a subscrip i on λ, to allow one to distinghuish between the λ-values of differewnt screws. A better sentence is therefore "…screw with axis $l_i = \{q_i + \lambda_i\omega_i : \lambda_i \in \mathbb{R}^1\} \ldots$"

On the expression for α just above eq. (2.72) p. 66:

This equation is based on the assumption that the two vectors ω_1 and ω_2 are both unit vectors. The cross product between unit vectors ω_1 and ω_2 is a vector in the direction of n with magnitude sin α. So, the dot product of this vector with the unit vector n is the scalar sin α. The dot product between unit vectors ω_1 and ω_2 is cos α. The equation is written in a cleaner way as

$$\alpha = \text{atan2}\left([\hat{\omega}_1\omega_2]^T n, \omega_1^T\omega_2 \right) \ .$$

The implication of this is also that, if cos α and sin α are needed to calculate the reciprocal product between two twists with eq. (2.72), one can simply calculate these with

$$\sin\alpha = [\hat{\omega}_1\omega_2]^T n \qquad ; \qquad \cos\alpha = \omega_1^T\omega_2$$

(Continued on next page.)

On the proof of Proposition 2.18, pp. 66, 67

The part of the proof for the case where both h_1 and h_2 are finite, given on p. 67, is re-written here and expanded a bit, to clarify. The twist V and wrench F are first re-written in a cleaner way:

$$V = M_1 \begin{bmatrix} \hat{q}_1\omega_1 + h_1\omega_1 \\ \omega_1 \end{bmatrix}$$

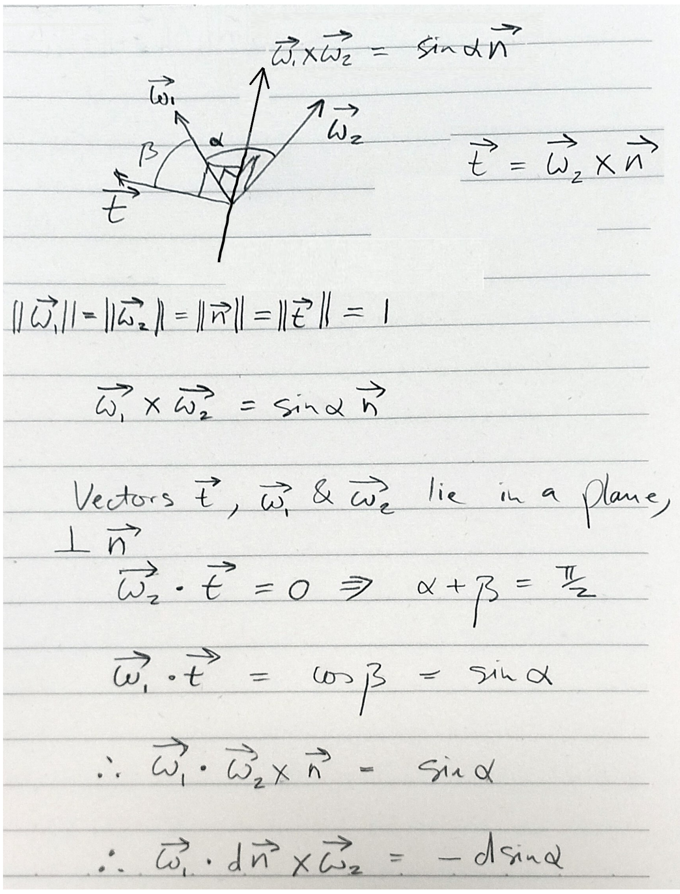
$$F = M_2 \begin{bmatrix} \omega_2 \\ \hat{q}_2\omega_2 + h_2\omega_2 \end{bmatrix}$$

Before looking at the instantaneous work between V and F, though, take note of the identity proven in the handdrawn sketch and handwriting in the figure to the right. The concluding equation can be written in component form as:

$$d\omega_1^T(\hat{n}\omega_2) = -d\,\text{sin}\,\alpha\ .$$

The expression for the intsantaneous work between V and F is then written in a cleaner form as:

$$V^TF = M_1M_2(\omega_2^T(\hat{q}_1\omega_1 + h_1\omega_1) + \omega_1^T(\hat{q}_2\omega_2 + h_2\omega_2))$$



Therefore,

$$\begin{aligned} V^TF &= M_1M_2\left\{\omega_2^T(\hat{q}_1\omega_1) + h_1\omega_1^T\omega_2 + \omega_1^T\left[(q_1 + dn)^\wedge\omega_2\right] + h_2\omega_1^T\omega_2\right\} \\ &= M_1M_2\left\{(h_1 + h_2)\,\text{cos}\,\alpha + d\omega_1^T(\hat{n}\omega_2) + \omega_2^T(\hat{q}_1\omega_1) + \omega_1^T(\hat{q}_1\omega_2)\right\} \end{aligned}$$

Now, the identity

$$\vec{\omega_2} \cdot \vec{q_1} \times \vec{\omega_1} + \vec{\omega_1} \cdot \vec{q_1} \times \vec{\omega_2} = 0$$

or

$$\omega_2^T(\hat{q}_1\omega_1) + \omega_1^T(\hat{q}_1\omega_2) = 0$$

is proven in the MATLAB script “dot_cross_ident2.pdf”, which is available on Moodle under the ”Notes” topic. This identity and the one from the handwritten proof are used to now conclude

$$V^TF = M_1M_2\left[(h_1 + h_2)\,\text{cos}\,\alpha - d\,\text{sin}\,\alpha\right]\ .$$

Looking at the unnumbered equation at the bottom of MLS p. 67,

$$V_1 \odot V_2 = v_1^T\omega_2 + v_2^T\omega_1$$

one can argue that $||\omega_1||$ and $||\omega_2||$ cannot be required to be 1 in general. This raises the question: are ω_1 and ω_2 in the proof of proposition 2.18 in the book and also discussed above, unit vectors?

It turns out that the formula for calculating α given on p. 66 is valid even if ω_1 and ω_2 are not unit vectors. To claim that

$$\text{sin}\,\alpha = \left[\hat{\omega}_1\omega_2\right]^T n \qquad ; \qquad \text{cos}\,\alpha = \omega_1^T\omega_2\ ,$$

as I have done above, though, is valid only if ω_1 and ω_2 are unit vectors.

Now in the proof of proposition 2.18, the writing of

$$F = M_2 \begin{bmatrix} \omega_2 \\ \hat{q}_2\omega_2 + h_2\omega_2 \end{bmatrix}$$

directly comes from the proof of Poinsot's theorem, Theorem 2.17, p. 65, and in this ω (or then ω_2 in this case) is clearly a unit vector: $\omega = f/||f||$.

In the case of writing the 6×1 velocity vector V, a twist, as a screw, it is clear that there cannot be any constraint that its bottom 3×1 sub-vector must be a unit vector, and the theory presented by MLS on pp. 46 to 48 does actually provide for the possibility that $||\omega|| \neq 1$. My interpretation is that, if one uses this theory to write a velocity twist as a screw and one then writes it as

$$V = M_1 \begin{bmatrix} \hat{q}_1\omega_1 + h_1\omega_1 \\ \omega_1 \end{bmatrix},$$

the actual angular velocity vector (the bottom 3×1 sub-vector) is $M_1\omega_1$ and the linear velocity vector (the upper 3×1 sub-vector) is $M_1(\hat{q}_1\omega_1 + h_1\omega_1)$, where M_1 is the magnitude of the angular velocity vector. The implication of this is that, in the proof of proposition 2.18, ω_1, like ω_2, is a unit vector. Also, the proof of proposition 2.18 actually makes use of the fact that both ω_1 and ω_2 are unit vectors.

This then leads to the further conclusion that the ω_1 and ω_2 in the unnumbered equation at the bottom of p. 67 cannot be interpreted as the same as ω_1 and ω_2 in the proof of proposition 2.18, as in the equation at the bottom of p. 67 the angular velocities in the velocity twist vectors cannot be limited to be unit vectors.